物理各科相關參考書目

—常用書籍

—聖經書籍

普通物理學

- Raymond A. Serway and John W. Jewett, *Physics for Scientists and Engineers*.
- David Halliday, Robert Resnick and Jearl Walker, *Fundamentals of Physics*.
- Richard P. Feynman, *Lectures on Physics, Vol. 1*.
- 大學物理學上、下冊，賴樹聲 編。(研究所考試用書)
- 大學物理學，共四冊，李怡嚴 著。(可加強觀念，難度較一般普物用教科書高)

數學、工程數學、應用數學及物理數學

向量及張量分析：

- Murray R. Spiegel, *Schaum's Outline of Vector Analysis and An Introduction to Tensor Analysis*.
- Eutiquio C. Young, *Vector and Tensor Analysis*.

線性代數：

- Howard Anton and Chris Rorres, *Elementary Linear Algebra with Applications*.(入門書籍)
- Steven J. Leon, *Linear Algebra with Applications*.
- Stephen H. Friedberg, Arnold J. Insel and Lawrence E. Spence, *Linear Algebra*.
- Ray Kunze and Kenneth Hoffman, *Linear Algebra*.
- 線性代數及其應用，黃子嘉 編。(考試用書)

特殊函數：

- E. T. Whittaker, G. N. Watson, *A Course of Modern Analysis (Cambridge Mathematical Library)*.
- 特殊函數概論，王竹溪、郭敦仁 著。(較難，不適合自修，可讀前二章，其餘當查閱用)

群論：

- Wu-Ki Tung, *Group Theory in Physics*.(不討論點群)
- Robert N. Cahn, *Semi-Simple Lie Algebras and their Representations*.(不討論點群)
- Howard Georgi, *Lie Algebras in Particle Physics*.

高等微積分、實變分析與數學分析：

- Walter Rudin, *Principles of Mathematical Analysis*.
- Walter Rudin, *Real and Complex Analysis*.
- Walter Rudin, *Functional Analysis*.
- Tom M. Apostol, *Mathematical Analysis*.
- Jerrold E. Marsden and Michael J. Hoffman, *Elementary Classical Analysis*.

幾何、微分幾何與黎曼幾何：

- Manfredo Perdigao do Carmo, *Differential Geometry of Curves and Surfaces*.(常用標準教科書)
- Manfredo Perdigao do Carmo and Francis Flaherty, *Riemannian Geometry*.
- Barrett O'Neill, *Elementary Differential Geometry*.(講法較新，內容廣泛)
- Barrett O'Neill, *Semi-Riemannian Geometry: With Applications to Relativity*.(需有微分幾何基礎，並以 Semi-Riemannian Geometry 嚴謹地討論狹義與廣義相對論)

工程數學：

- Erwin Kreyszig, *Advanced Engineering Mathematics*.（討論較多物理例題與觀念）
- Peter V. O'Neil, *Advanced Engineering Mathematics*.（講述較多數學的概念和方法）
- 工程數學學習要訣，劉明昌 著。(講解清晰，適合自修)
- 高等工程數學，程雋 著。(內容、觀念較多，收錄眾多國內研究所工數考古題，內容及排版有點像教科書)

應用數學及物理數學：

- Hans J. Weber and George B. Arfken, *Essential Mathematical Methods for Physicists*.（特別為大學部學生程度編寫，較為原版本 Arfken 精簡）
- George B. Arfken and Hans J. Weber, *Mathematical Methods for Physicists*.（一般物理系應用數學常用書）
- Mary L. Boas, *Mathematical Methods in the Physical Sciences*.（內容講解清晰，亦適合自修，近幾年越來越多人使用）
- Bruce R Kusse and Erik A. Westwig, *Mathematical Physics : Applied Mathematics for Scientists and Engineers*.
- Harold Jeffreys and Bertha Jeffreys, *Methods of Mathematical Physics (Cambridge Mathematical Library)*.
- Mathews and Walker, *Mathematical Methods for Physicists*.
- K. F. Riley, M. P. Hobson, S. J. Bence, *Mathematical Methods for Physics and Engineering: A Comprehensive Guide*.（內容較 Arfken 廣泛且詳細，是不錯的參考書籍）
- I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*.
- Peter Szekeres, *A Course in Modern Mathematical Physics: Groups, Hilbert Space and Differential Geometry*.(不同於一般物理數學教科書，著重在近代理論物理需要的數學工具)
- R. Courant and D. Hilbert, *Methods of Mathematical Physics (2 vol. set)*.
- Michael Reed and Barry Simon, *Methods of modern Mathematical Physics, vol. 1: Functional Analysis*.
- Michael Reed and Barry Simon, *Methods of modern Mathematical Physics, vol. 2: Fourier Analysis, Self-Adjointness*.
- Michael Reed and Barry Simon, *Methods of modern Mathematical Physics, vol. 3: Scattering Theory*.
- Michael Reed and Barry Simon, *Methods of modern Mathematical Physics, vol. 4: Analysis of operators*.
- Bernard F. Schutz, *Geometrical Methods of Mathematical Physics*.（較偏重數學，講述幾何在物理上的應用）
- Theodore Frankel, *The Geometry of Physics: An Introduction*.（清楚地講述幾何與物理的關係以及物理意義）
- Mikio Nakahara, *Geometry, Topology and Physics*.（由簡明的量子場論開始，然後講述拓樸、幾何與物理的關係和應用，亦簡略地介紹了弦理論）
- 數學物理方法，梁昆淼 著。
- 數學物理方法，郭敦仁 著。

非交換幾何：

- Alain Connes, *Noncommutative Geometry*.(作者與 Carlo Rovelli 基於 thermal time hypothesis 制定了量子場論的協變模型)
- Alain Connes and Matilde Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*.

理論力學、分析力學及古典力學

- W. Greiner, *Classical Mechanics: Point Particles and Relativity*.
- W. Greiner, *Classical Mechanics: System of Particles and Hamiltonian Dynamics*.
- Fowles and Cassiday, *Analytical Mechanics*.（講述較 Marion 好但內容較少）

- Stephen T. Thornton and Jerry B. Marion, *Classical Dynamics of Particles and Systems*.
- Arnold Sommerfeld, *Lectures on Theoretical Physics: Mechanics (Volume I)*.
- L. D. Landau and E. M. Lifshitz, *Mechanics (Course of Theoretical Physics, Volume 1)*.
- L D Landau, E.M. Lifshitz, L. P. Pitaevskii, and A. M. Kosevich, *Theory of Elasticity (Course of Theoretical Physics, Volume 7)*.
- Herbert Goldstein, Charles P. Poole and John L. Safko, *Classical Mechanics*.
- V. I. Arnold, A. Weinstein, and K. Vogtmann, *Mathematical Methods of Classical Mechanics*.（較偏重數學，以現代數學嚴謹地討論，講述微分幾何的基礎以及幾何與古典力學的關係）
- 理論力學教程，周衍柏 著。
- 理論物理第一冊 古典動力學，吳大猷 著。(Lagrange 及 Hamilton 動力學)

流體力學

- L. D. Landau and E.M. Lifshitz, *Fluid Mechanics (Course of Theoretical Physics, Volume 6)*.

工程電磁學、微波工程、電磁學及電動力學

工程電磁學與微波工程：

- David K. Cheng, *Field and Wave Electromagnetics*.（一般工學院電磁學常用書）
- Nannapaneni Narayana Rao, *Elements of Engineering Electromagnetics*.(內容較多且偏重應用，光電類常用書)
- Robert E. Collin, *Foundations for Microwave Engineering*.
- David M. Pozar, *Microwave Engineering*.

電磁學及電動力學：

- E. M. Purcell, *Electricity and Magnetism (Berkeley Physics, Volume 2)*.（只講述靜電和靜磁學，觀念清楚明瞭）
- Richard P. Feynman, *Lectures on Physics, Vol. 2*.
- David J. Griffiths, *Introduction to Electrodynamics*.（物理系電磁學常用書）
- W. Greiner, *Classical Electrodynamics*.
- Mark A. Heald and Jerry B. Marion, *Classical Electromagnetic Radiation*.(難度介於 Griffiths 與 Jackson 之間，講解清晰)
- L. D. Landau and E.M. Lifshitz, *The Classical Theory of Fields (Course of Theoretical Physics, Volume 2)*.
- L. D. Landau, E.M. Lifshitz, and L. P. Pitaevskii, *Electrodynamics of Continuous Media (Course of Theoretical Physics, Volume 8)*.
- John D. Jackson, *Classical Electrodynamics*.（建議先唸完 Griffiths 的電動力學導論，頗有幫助）
- 電動力學，郭碩鴻 著。
- 理論物理第三冊 電磁學，吳大猷 著。(大二程度至 Jackson 程度之前)

熱力學、熱物理、統計熱物理及統計力學

熱力學：

- M. W. Zemansky and Richard H. Dittman, *Heat and Thermodynamics*.
- 熱力學，王竹溪 著。

熱物理、統計熱物理及統計力學：

- Charles Kittel, *Thermal Physics*.（以統計觀念及方法著手講述熱統計物裡）

- Ashley H. Carter, *Classical and Statistical Thermodynamics*.（前半部討論熱力學，後半部開始討論統計物裡）
- Kerson Huang, *Introduction to Statistical Physics*.（物理討論清晰，並有討論些較特別的 topics）
- W. Greiner, *Thermodynamics and Statistical Mechanics*.
- Frederick Reif, *Fundamentals of Statistical and Thermal Physics*.（較早的統計物理經典書籍）
- E. M. Lifshitz and L. D. Landau, *Statistical Physics, Part 1 (Course of Theoretical Physics, Volume 5)*.
- E.M. Lifshitz and L. P. Pitaevskii, *Statistical Physics, Part 2 (Course of Theoretical Physics, Volume 9)*.
- E.M. Lifshitz and L. P. Pitaevskii, *Physical Kinetics (Course of Theoretical Physics, Volume 10)*.
- R. K. Pathria, *Statistical Mechanics*.
- Kerson Huang, *Statistical Mechanics*.
- 熱力學、統計物理，汪志誠 著。(未涉及密度矩陣，為學習量子物理前的統計物理用書)
- 統計物理學導論，王竹溪 著。(同樣為學習量子物理前的統計物理用書，討論少許量子統計物理)
- 統計力學，李政道 著。
- 理論物理第五冊 熱力學，氣體運動論，統計力學，吳大猷 著。

近代物理學、量子物理學及量子力學

近代物理學（量子物理學與基礎狹義相對論）：

- Ralph Llewellyn and Paul A. Tipler, *Modern Physics*.
- Robert Martin Eisberg, *Fundamentals of Modern Physics*.（數學推導較多，為 Eisberg 所寫 *Quantum Physics* 的舊版，較適合想多了解量子力學中 Schrödinger 波動力學的人，有些不錯的討論，但內容較老舊）
- Robert Eisberg and Robert Resnick, *Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles*.（量子物理的觀念說明較多）
- Arthur Beiser, *Concepts of Modern Physics*.（電子及電機系常用書，台大光電用書）
- 理論物理第二冊 量子論與原子結構，吳大猷 著。(古量子論及光譜學)

量子物理學及量子力學：

- A.P. French and Edwin F. Taylor, *An Introduction to Quantum Physics (MIT Introductory Physics Series)*.
- Richard P. Feynman, *Lectures on Physics, Vol. 3*.
- David J. Griffiths, *Introduction to Quantum Mechanics*.（難度適中，內容較 Gasiorowicz 少）
- Stephen Gasiorowicz, *Quantum Physics*.（簡明扼要，省略較多數學推導，但觀念講述很多，補充資料亦有 WKB Approximation, A-B Effect, Interaction Picture and Field Quantization 等資料可至 Wesley 網站下載）
- Claude Cohen-Tannoudji, Bernard Diu and Frank Laloe, *Quantum Mechanics (2 vol. set)*.
- L. D. Landau and E. M. Lifshitz, *Quantum Mechanics Non-Relativistic Theory (Course of Theoretical Physics, Volume 3)*.
- W. Greiner, *Quantum Mechanics: An Iintroduction*.
- W. Greiner, *Quantum Mechanics: Special Chapters*.
- W. Greiner and B. Müller, *Quantum Mechanics: Symmetries*.
- W. Greiner, *Relativistic Quantum Mechanics-Wave Equations*.
- Eugen Merzbacher, *Quantum Mechanics*.（內容的排版可分二部分，前半部分程度接近大學部量力程度像量物教科書；後半部講述大學時未講述部分內容）
- Richard P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals*.
- J. J. Sakurai, *Modern Quantum Mechanics, Revised Edition*.（簡明扼要，較難，物理現象及觀念說明較多，也

較需要思考，建議先唸完 Griffiths 的量子力學導論）

- R. Shankar, *Principles of Quantum Mechanics*.（數學及說明清晰，不難懂，亦適合初學者，包括少許相對論量子力學的討論）
- P. A. M. Dirac, *The Principles of Quantum Mechanics*.（經典量子力學書籍）
- 量子力學導論，曾謹言 著。
- 量子力學，張永德 著。
- 量子力學 卷一，曾謹言 著。(內容與量子力學導論差不多，但討論較多較詳細)
- 量子力學 卷二，曾謹言 著。(深入討論量子力學、相對論量子力學及輻射場量子化問題)
- 量子力學中的角動量，林辛未 著。(討論還算清晰)
- 高等量子力學，喀興林 著。(內容嚴謹)
- 高等量子力學 上下冊，張永德 著。(內容包含非相對論性及相對論性量子場論，並有量子化問題的討論)
- 理論物理第六冊 量子力學（甲部），吳大猷 著。(矩陣力學及波動力學)
- 理論物理第七冊 量子力學（乙部），吳大猷 著。(相對論量子力學及量子場論)

註：

在 Stephen Gasiorowicz 的 *Quantum Physics* 及櫻井 純 (J. J. Sakurai) 的 Modern Quantum Mechanics 的最後有些量子力學書籍的介紹及清楚的說明。

固態物理學及凝聚態物理學

- Charles Kittel, *Introduction to Solid State Physics*.
- Charles Kittel, *Quantum Theory of Solids*.(屬高等固態物理)
- Neil W. Ashcroft and N. David Mermin, *Solid State Physics*.
- Giuseppe Grosso and Giuseppe Pastori Parravicini, *Solid State Physics*.（講述方法較新）
- Michael P. Marder, *Condensed Matter Physics*.
- L D Landau, E.M. Lifshitz, L. P. Pitaevskii, and A. M. Kosevich, *Theory of Elasticity (Course of Theoretical Physics, Volume 7)*.
- E.M. Lifshitz and L. P. Pitaevskii, *Physical Kinetics (Course of Theoretical Physics, Volume 10)*.
- Philip L. Taylor and Olle Heinonen, *A Quantum Approach to Condensed Matter Physics*.
- 固體物理學，黃昆 著。(盡量不用較深的數學方法，討論易懂)

超導物理學

- James F. Annett, *Superconductivity, Superfluids, and Condensates (Oxford Master Series in Condensed Matter Physics)*.
- Michael Tinkham, *Introduction to Superconductivity*.
- Charles P. Poole Jr., Horacio A. Farach, Richard J. Creswick, and Ruslan Prozorov, *Superconductivity*.

凝態物理的量子場論(非相對論性量子場論)、多體物理與強關聯電子系統

- Charles Kittel, *Quantum Theory of Solids*.(屬高等固態物理)
- Naoto Nagaosa, *Quantum Field Theory in Condensed Matter Physics*.
- Naoto Nagaosa, *Quantum Field Theory in Strongly Correlated Electronic Systems*.
- Alexander Altland, Ben D. Simons, *Condensed Matter Field Theory*.

- Alexei M. Tsvelik, *Quantum field theory in condensed matter physics*.
- Gabriele Giuliani, Giovanni Vignale, *Quantum Theory of the Electron Liquid*.
- Henrik Bruus and Karsten Flensberg, *Many-body Quantum Theory in Condensed Matter Physics: An Introduction*.
- Gerald D. Mahan, *Many-Particle Physics*.
- Philippe A. Martin, Francois Rothen, *Many-Body Problems and Quantum Field Theory: An Introduction*.
- Xiao-Gang Wen, *Quantum Field Theory of Many-body Systems: From the Origin of Sound to an Origin of Light and Electrons*.

光學

- Frank L Pedrotti, Leno M Pedrotti, and Leno S Pedrotti, *Introduction to Optics*.（內容較 Hecht 少）
- Eugene Hecht, *Optics*.（常用教科書）
- Joseph W. Goodman, *Introduction to Fourier Optics*.
- Max Born and Emil Wolf, *Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light*.
- 光學，趙凱華 著。

雷射物理

- Anthony E. Siegman, *Lasers*.

非線性動力學

- Robert Hilborn, *Chaos and Nonlinear Dynamics: An Introduction for Scientists and Engineers*.
- Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry and Engineering*.

原子物理學

- Christopher J. Foot, *Atomic Physics (Oxford Master Series in Atomic, Optical and Laser Physics)*.

X 光繞射學

- B.D. Cullity and S.R. Stock, *Elements of X-Ray Diffraction*.

軟物質

- Richard A.L. Jones, *Soft Condensed Matter (Oxford Master Series in Condensed Matter Physics)*.

計算物理學

- Rubin H. Landau, Manuel J. Páez, and Cristian C. Bordeianu, *Computational Physics: Problem Solving with Computers*.

微電子電路學

- Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits (The Oxford Series in Electrical and Computer Engineering)*.

半導體物理

- Simon M. Sze and Kwok K. Ng, *Physics of Semiconductor Devices*.
- Simon M. Sze, *Semiconductor Devices: Physics and Technology*.
- Donald A. Neamen, *Semiconductor Physics and Devices*.

光電子學

- Safa O. Kasap, *Optoelectronics and Photonics: Principles and Practices*.（較簡單，內容較少）
- Bahaa E. A. Saleh and Malvin Carl Teich, *Fundamentals of Photonics*.
- Amnon Yariv and Pochi Yeh, *Photonics: Optical Electronics in Modern Communications*.

介觀物理

- Supriyo Datta, *Electronic Transport in Mesoscopic Systems*.（入門書籍）

加速器物理

- Klaus Wille and Jason McFall, *The Physics of Particle Accelerators: An Introduction*.
- E. J. N. Wilson, *An Introduction to Particle Accelerators*.
- D. A. Edwards and M. J. Syphers, *An Introduction to the Physics of High Energy Accelerators*.
- S. Y. Lee, *Accelerator Physics*.
- Helmut Wiedemann, *Particle Accelerator Physics*.

輻射偵測與度量

- Nicholas Tsoulfanidis, *Measurement and Detection of Radiation*.
- Glenn F. Knoll, *Radiation Detection and Measurement*.

核物理、粒子物理學（高能物理學）與粒子天文物理學

- W. Greiner and J.A. Maruhn, *Nuclear Models*.
- David Halliday, *Introductory Nuclear Physics*.（較早期的經典書籍）
- Kenneth S. Krane, *Introductory Nuclear Physics*.（為 Halliday 核物導論的修正及擴充）
- Richard C. Fernow, *Introduction to Experimental Particle Physics*.（實驗粒子物理）
- Donald H. Perkins, *Introduction to High Energy Physics*.
- Donald H. Perkins, *Particle Astrophysics (Oxford Master Series in Particle Physics, Astrophysics, and Cosmology)*.
- G. L. Kane and Gordon Kane, *Modern Elementary Particle Physics: Updated Edition*.
- David J. Griffiths, *Introduction to Elementary Particles*.
- John Dirk Walecka, *Theoretical Nuclear and Subnuclear Physics*.（以核物理及強子物理為主）
- F. Halzen and A. D. Martin, *Quarks and Leptons: An Introductory Course in Modern Particle Physics*.
- Kerson Huang, *Quarks, Leptons & Gauge Fields*.
- Ta-Pei Cheng and Ling-Fong Li, *Gauge Theory of Elementary Particle Physics*.
- Ta-Pei Cheng and Ling-Fong Li, *Gauge Theory of Elementary Particle Physics: Problems and Solutions*.

- Felix Boehm and Petr Vogel, *Physics of Massive Neutrinos*.
- Masataka Fukugita and Atsuto Suzuki, *Physics and Astrophysics of Neutrinos*.（較舊，但後半部講述了很多與 neutrino 實驗相關的 topics）
- Masataka Fukugita and Tsutomu Yanagida, *Physics of Neutrinos: and Applications to Astrophysics*.（較新，內容為擴充了 *Physics and Astrophysics of Neutrinos* 前 200 頁 neutrino 物理理論的部分）
- Carlo Giunti and Chung W Kim, *Fundamentals of Neutrino Physics And Astrophysics*.（由 Dirac equation 開始講述量子場論，進而討論至 neutrino physics，內容相較其他 neutrino 書籍而言有很多較詳細的討論）
- W. N. Cottingham and D. A. Greenwood, *An Introduction to the Standard Model of Particle Physics*.
- 粒子物理與核物理講座，高崇壽、曾謹言 著。
- 粒子物理和場論，李政道 著。

實驗物理中的機率與統計方法

- Byron P. Roe, *Probability and Statistics in Experimental Physics (Undergraduate Texts in Contemporary Physics)*.
- Louis Lyons, *Statistics for Nuclear and Particle Physics*.
- Glen Cowan, *Statistical Data Analysis*.
- Frederick James, *Statistical Methods in Experimental Physics*.
- 實驗物理中的機率和統計，朱永生 著。

天文學與天文物理學

- Zeilik, Michael, and Stephen A. Gregory, *Introductory Astronomy and Astrophysics*.
- Bradley W. Carroll and Dale A. Ostlie, *An Introduction to Modern Astrophysics*.

狹義相對論、廣義相對論與引力理論

- P. French, *Special Relativity (MIT Introductory Physics)*.
- Ta-Pei Cheng, *Relativity, Gravitation and Cosmology: A Basic Introduction (Oxford Master Series in Particle Physics, Astrophysics, and Cosmology)*.（論述清晰，前面以物理為主，後面再討論相對論中的數學）
- Bernard F. Schutz, *A First Course in General Relativity*.（常用教科書）
- James B. Hartle, *Gravity: An Introduction to Einstein's General Relativity*.（常用教科書，以相對論的物理為主來出發講述）
- Sean Carroll, *Spacetime and Geometry: An Introduction to General Relativity*.（常用教科書，內容包含了較新較前沿的引力理論簡介，並提及了引力理論該有的所有基礎知識，最後一章亦簡介了彎曲時空的量子場論，是非常好的教科書之一）
- L. D. Landau and E.M. Lifshitz, *The Classical Theory of Fields (Course of Theoretical Physics, Volume 2)*.
- James J. Callahan, *The Geometry of Spacetime: An Introduction to Special and General Relativity*.（數學成分重，全書以講述幾何的方式詳細並非常清楚地講述相對論，特別以時空圖來講解狹義相對論）
- Theodore Frankel, *Gravitational Curvature: An Introduction to Einstein's Theory*.
- P. A. M. Dirac, *General Theory of Relativity*.（以簡單清晰的方式講述相對論，為常用參考書籍）
- Albert A. Einstein, *Relativity: The Special and the General Theory*.
- Richard Feynman, Fernando Morinigo, William Wagner, and Brian Hatfield, *Feynman Lectures on Gravitation*.

（由電磁學出發討論自旋 2 重力子，以場論方式推導出愛因斯坦方程式，不同一般相對論書籍以幾何觀點來討論）

- Steven Weinberg, *Gravitation and Cosmology*.（經典的重力學與宇宙學用書）
- Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler, *Gravitation*.（經典的重力學用書）
- Robert M. Wald, *General Relativity*.（系統地討論相對論中的重要課題，目前為標準參考用書）
- S. W. Hawking, G. F. R. Ellis, *The Large Scale Structure of Space-Time*.
- John D. Stewart, *Advanced General Relativity (Cambridge Monographs on Mathematical Physics)*.
- Yasunori Fujii and Kei-ichi Maeda, *The Scalar-Tensor Theory of Gravitation (Cambridge Monographs on Mathematical Physics)*.
- Valerio Faraoni, *Cosmology in Scalar-Tensor Gravity (Fundamental Theories of Physics)*.
- Barrett O'Neill, *Semi-Riemannian Geometry: With Applications to Relativity*.（需有微分幾何基礎，並以 Semi-Riemannian Geometry 嚴謹地討論狹義與廣義相對論）
- Tian Yu Ca, *Conceptual Developments of 20th Century Field Theories*.（專門而有深度地討論了古典場論、量子場論以及至規範場論的哲學書籍）
- 理論物理第四冊 相對論，吳大猷 著。(狹義及廣義相對論)

黑洞物理

- Derek Raine, *Black Holes: An Introduction*.
- Barrett O'Neill, *Geometry of Kerr Black Holes*.（數學成分重）

宇宙學

- Steven Weinberg, *The First Three Minutes: A Modern View Of The Origin Of The Universe*.（學習宇宙學必讀書籍，為著名宇宙學科普書）
- Andrew Liddle, *Introduction to modern cosmology*.(內容不多，提及了幾乎所有宇宙學重要基礎的入門書籍，適合自修)
- Scott Dodelson, *Modern Cosmology*.（近幾年常用書籍，有系統地描述微擾理論）
- Viatcheslav Mukhanov, *Physical Foundations of Cosmology*.（內容豐富，以基本原理為出發講述）
- Steven Weinberg, *Cosmology*.（有詳細地數學推導，某些 topics 及細節需查閱作者所寫的 *Gravitation and Cosmology* 及 *The Quantum Theory of Fields*）
- J. A. Peacock, *Cosmological Physics*.
- Gerhard Börner, *The Early Universe*.（資料較新，內容亦廣泛）
- Phillip James Edwin Peebles, *Principles of Physical Cosmology*.
- Phillip James Edwin Peebles, *Large-Scale Structure of the Universe*.
- Edward Kolb and Michael Turner , *The Early Universe: REPRINTS*.（作者所編輯的相關期刊論文集）
- Edward Kolb and Michael Turner , *The Early Universe*.（資料較舊，標準參考書籍）
- A. D. Linde, *Particle Physics and Inflationary Cosmology (Contemporary Concepts in Physics) (vol 5)*, arXiv:0503203v1 [hep-th].
- Andrew R. Liddle and David H. Lyth, *Cosmological Inflation and Large-Scale Structure*.
- David H. Lyth and Andrew R. Liddle, *The Primordial Density Perturbation: Cosmology, Inflation and the Origin of Structure*.（相較於第一版 *Cosmological Inflation and Large-Scale Structure* 而言，多了 non-Gaussianity 的

詳細討論，大尺度結構標準參考書籍）

- Ruth Durrer, *The Cosmic Microwave Background*.
- Valerio Faraoni, *Cosmology in Scalar-Tensor Gravity (Fundamental Theories of Physics)*.
- Salvatore Capozziello and Valerio Faraoni, *Beyond Einstein Gravity: A Survey of Gravitational Theories for Cosmology and Astrophysics*.（專門討超越廣義相對論的 modified gravity 的書籍，內容有很多詳細的討論）
- Bertone Gianfranco, *Particle Dark Matter: Observations, Models and Searches*.（為論文集，內容為目前最前沿且統整過的暗物質理論、觀測實驗以及理論模擬文章）
- Luca Amendola and Shinji Tsujikawa, *Dark Energy: Theory and Observations*.（目前最詳盡的暗能量理論書籍，包含 modified matter 與 modified gravity 理論以及非均勻宇宙模型）

相對論性量子場論

- Tian Yu Ca, *Conceptual Developments of 20th Century Field Theories*.（專門而有深度地討論了古典場論、量子場論至規範場論的哲學書籍）
- Amitabha Lahiri and Palash B. Pal, *A First Book of Quantum Field Theory*.（入門書籍，簡單清楚，適合自修，內容以 QED 為主）
- Michele Maggiore, *A Modern Introduction to Quantum Field Theory (Oxford Master Series in Statistical, Computational, and Theoretical Physics)*.（很好的入門書籍，亦適合自修）
- W. Greiner and J. Reinhardt, *Field Quantization*.
- W. Greiner and J. Reinhardt, *Quantum Electrodynamics*.
- W. Greiner, S. Schramma and E. Stein, *Quantum Chromodynamics*.
- W. Greiner and B. Müller, *Gauge Theory of Weak Interactions*.
- Paul A. M. Dirac, *Lectures on Quantum Mechanics*.（討論 Hamiltonian formalism、量子化問題、彎曲空間與平空間的量子化）
- V B Berestetskii, L. P. Pitaevskii, and E.M. Lifshitz, *Quantum Electrodynamics (Course of Theoretical Physics, Volume 4)*.
- T. Muta, *Foundations of Quantum Chromodynamics: An Introduction to Perturbative Methods in Gauge Theories (World Scientific Lecture Notes in Physics)*.（經典的 QCD 教科書）
- J. J. Sakurai, *Advanced Quantum Mechanics*.（內容為相對論量子力學與場論，全書以 covariant vector 表達）
- Lewis H. Ryder, *Quantum Field Theory*.（常用的場論入門書）
- J. D. Bjorken and S. D. Drell, *Relativistic Quantum Mechanics*.（描述詳細，但內容較老舊）
- J. D. Bjorken and S. D. Drell, *Relativistic Quantum Fields*.（描述詳細，但內容較老舊）
- Claude Itzykson and Jean-Bernard Zuber, *Quantum Field Theory*.（內容充實詳細，但應用較少）
- Michio Kaku, *Quantum Field Theory: A Modern Introduction*.（講述及整理清晰）
- Mark Srednicki, *Quantum Field Theory*.（近幾年大家常用書籍，內容廣泛，編排邏輯適當，但屬入門書籍，某些詳細 details 與 topics 需參考 Peskin 及 Weinberg）
- Brian Hatfield, *Quantum Field Theory of Point Particles and Strings*.
- Pierre Ramond, *Field Theory: A Modern Primer*.（以路徑積分為出發點，討論詳盡，亦討論重力的規範理論）
- A. Zee, *Quantum Field Theory in a Nutshell*.（由路徑積分出發，講法生動活潑且簡潔明瞭，有章節特別討論場論在凝態應用的教科書）
- Michael E. Peskin and Daniel V. Schroeder, *An Introduction to Quantum Field Theory*.（常用教科書）

- Steven Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*.（較嚴謹詳細且完整，較難，較不適合初學者）
- Steven Weinberg, *The Quantum Theory of Fields, Vol. II: Modern Applications*.
- Steven Weinberg, *The Quantum Theory of Fields, Vol. III: Supersymmetry*.
- Bryce S. DeWitt, *The Global Approach to Quantum Field Theory: 2-Volume Set (The International Series of Monographs on Physics, 114)*.
- Kerson Huang, *Quantum Field Theory : From Operators to Path Integrals*.（有很多不錯的討論，以 QED 及量子化為主，不討論 QCD 與 Electroweak Theory）
- Kerson Huang, *Quarks, Leptons & Gauge Fields*.
- Ta-Pei Cheng and Ling-Fong Li, *Gauge Theory of Elementary Particle Physics*.
- Ta-Pei Cheng and Ling-Fong Li, *Gauge Theory of Elementary Particle Physics: Problems and Solutions*.
- Stefan Pokorski, *Gauge Field Theories (Cambridge Monographs on Mathematical Physics)*.
- John C. Collins, *Renormalization: An Introduction to Renormalization, the Renormalization Group and the Operator-Product Expansion (Cambridge Monographs on Mathematical Physics)*.
- Jürgen Jost, *Geometry and Physics*.（數學家所寫，以數學的觀點來描述，但不特別強調數學的嚴謹性，前半部講述幾何與物理，後半部講述量子場論）
- Reinhold A. Bertlmann, *Anomalies in Quantum Field Theory (International Series of Monographs on Physics)*.
- Alain Connes and Matilde Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*.

彎曲時空的量子場論(半古典量子重力)

- Paul A. M. Dirac, *Lectures on Quantum Mechanics*.（討論 Hamiltonian formalism、量子化問題、彎曲空間與平空間的量子化）
- Viatcheslav Mukhanov and Sergei Winitzki, *Introduction to Quantum Effects in Gravity*.（入門書籍，僅討論 scalar field，不討論 spinor field 在 curved spacetime 的問題）
- Bryce S. DeWitt, *Quantum Field Theory in Curved Spacetime, Physics Reports 19, no. 6(1975), 295-357*.
- N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space (Cambridge Monographs on Mathematical Physics)*.（經典參考書籍）
- Stephen A. Fulling, *Aspects of Quantum Field Theory in Curved Spacetime (London Mathematical Society Student Texts)*.
- Robert M. Wald, *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics (Chicago Lectures in Physics)*.（論述簡潔，省略推導）
- Leonard Parker and David Toms, *Quantum Field Theory in Curved Spacetime: Quantized Fields and Gravity (Cambridge Monographs on Mathematical Physics)*.（推導詳細，包含詳細 spinor 和多維度 Dirac matrices 及其 representation 的討論）

超對稱理論與超引力理論

- Fred Cooper, Avinash Khare, and Uday Sukhatme, *Supersymmetry in Quantum Mechanics*.
- Stephen P. Martin, *A Supersymmetry Primer, e-print available as* [hep-ph/9709356].（預備知識為標準模型與量子場論）
- P. West, *Introduction to Supersymmetry and Supergravity*.

- Julius Wess, Jonathan Bagger, *Supersymmetry and Supergravity*.
- Ian J. R. Aitchison, *Supersymmetry in Particle Physics: An Elementary Introduction*.
- Howard Baer, Xerxes Tata, *Weak Scale Supersymmetry: From Superfields to Scattering Events*.
- Manuel Drees, Rohini Godbole, Probir Roy, *Theory and Phenomenology of Sparticles: An Account of Four-Dimensional N=1 Supersymmetry in High Energy Physics*.
- John Terning, *Modern Supersymmetry: Dynamics and Duality*.

量子重力(含迴圈量子重力)

- Lee Smolin, *Three Roads to Quantum Gravity*.（科普量子重力書籍）
- Carlo Rovelli, *A Dialog on Quantum Gravity, e-print available as* [hep-th/0310077].
- Abhay Ashtekar, *Gravity and the quantum, e-print available as* [gr-qc/0410054].
- Ashtekar, Abhay; Lewandowski, *Background Independent Quantum Gravity: A Status Report, e-print available as* [gr-qc/0404018].
- Steven Carlip, *Quantum Gravity in 2+1 Dimensions (Cambridge Monographs on Mathematical Physics)*.
- Carlo Rovelli, *Quantum Gravity (Cambridge Monographs on Mathematical Physics)*.（內容講述迴圈量子重力）
- Thomas Thiemann, *Modern Canonical Quantum General Relativity (Cambridge Monographs on Mathematical Physics)*.
- Claus Kiefer, *Quantum Gravity*.
- Alain Connes and Matilde Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*.

弦理論與 M 理論

- Brian Greene, *The Elegant Universe: Superstrings, Hidden Dimensions, and the Quest for the Ultimate Theory*.（講述弦論概念而不牽涉艱深微分幾何數學的弦論科普書籍）
- Barton Zwiebach, *A First Course in String Theory*.（為大學部學生而寫的弦論書籍，內容亦包括了最新發展的 AdS/CFT 理論）
- Brain Hatfield, *Quantum Field Theory of Particles and Strings*.
- Katrin Becker, Melanie Becker and John H. Schwarz, *String Theory and M-Theory: A Modern Introduction*.
- Joseph Polchinski, *String Theory, Vol. 1: An Introduction to the Bosonic String (Cambridge Monographs on Mathematical Physics)*.
- Joseph Polchinski, *String Theory, Vol. 2: Superstring Theory and Beyond (Cambridge Monographs on Mathematical Physics)*.
- Michael B. Green, John H. Schwarz, and Edward Witten, *Superstring Theory, Vol. 1: Introduction (Cambridge Monographs on Mathematical Physics)*.
- Michael B. Green, John H. Schwarz, and Edward Witten, *Superstring Theory, Vol. 2: Loop amplitudes, anomalies and phenomenology (Cambridge Monographs on Mathematical Physics)*.

理論物理

- Arnold Sommerfeld, *Lectures on Theoretical Physics (6 vol. set)*.（年代較久遠）
- E. M. Lifshitz and L. D. Landau etc., *Course of Theoretical Physics (10 vol. set)*.
- W. Greiner, *Theoretical Physics (13 vol. set)*.

■ 理論物理 七冊，吳大猷 著。

註：

其他書籍以及其細節與討論請參考台灣大學 BBS 批踢踢實業坊《Physics》物理版置底文—自修書及推薦區或 Wikipedia 網站。

其他—程式語言

C 語言：

■ Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language (2nd Edition)*.

C++語言：

■ Stanley B. Lippman, Josée Lajoie and Barbara E. Moo, *C++ Primer (4th Edition)*.

國立屏東教育大學 應用物理系

96 級畢業生 羅令歲 編輯於 2007.04.09

Jason Ling-Wei Luo @ Department of Physics, National Tsing Hua University

Re-edited on Oct 08, 2011